QIntern 2026 Project Proposal

Project Title: Quantum-Enhanced Neural ODEs and QAOA for Dynamic Energy System Optimization

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Objectives

The primary goal of this project is to upgrade classical deep learning and operations research models used in dynamic energy management systems by integrating Quantum Machine Learning (QML) and quantum optimization algorithms. Specifically, the project replaces traditional Mixed-Integer Linear Programming (MILP) with Quantum Approximate Optimization Algorithm (QAOA) and enhances differential equation-based forecasting with Quantum Neural ODEs.

Specific objectives include:

- Formulating industrial energy load-shifting and cost minimization problems as Quadratic Unconstrained Binary Optimization (QUBO) models to be solved using QAOA.
- Integrating Parameterized Quantum Circuits (PQCs) into Physics-Informed Neural Networks (Q-PINNs) and continuous-time Neural ODEs.
- Designing robust quantum embedding strategies to encode high-dimensional, stochastic time-series energy datasets into quantum states.
- Benchmarking the hybrid quantum-classical pipeline against classical baselines, **with experimental explorations into Quantum Reinforcement Learning and real hardware deployments.**

Work Plan (1st July - 15 August)

Week 1 (July 1 - July 7): Literature Review & Quantum Formulation

- Review recent literature on Quantum Neural ODEs, Q-PINNs, and QAOA applied to time-series forecasting and smart grids.
- Set up the quantum simulation frameworks (e.g., Qiskit, PennyLane) and define the QUBO mathematical formulation for the energy arbitage problem.

Week 2 (July 8 - July 14): Data Encoding & Preprocessing

- Preprocess large-scale grid and IoT consumption datasets.
- Experiment with various quantum feature maps (Angle, Amplitude, and Hamiltonian embeddings) to efficiently load continuous-time energy data into quantum circuits.

Week 3 (July 15 - July 21): QML Architecture Development

- Develop and train the Quantum Neural ODEs and Q-PINNs.
- Embed physical constraints (e.g., grid capacity, thermal dissipation) into the quantum loss function to ensure thermodynamically feasible predictions.

Week 4 (July 22 - July 28): QAOA Implementation & QRL Exploration

- Implement QAOA to solve the formulated QUBO problem for scheduling dynamic industrial loads.

- **Experimental Phase (Quantum Reinforcement Learning): Formulate the load-shifting decision process as a Markov Decision Process and test a prototype Quantum Deep Q-Network (Q-DQN) to compare dynamic policy learning against static QAOA optimization.**

Week 5 (July 29 - August 4): Hardware Testing & Error Mitigation

- Benchmark the fully integrated quantum pipeline against classical Deep Learning and MILP approaches in simulated NISQ environments.
- **Experimental Phase (Real Hardware Deployment): Execute the optimized quantum circuits on real quantum backends (e.g., IBM Quantum processors). Apply Error Mitigation techniques (such as Zero-Noise Extrapolation - ZNE) to analyze and reduce the impact of hardware noise on the optimization results.**

Week 6 (August 5 - August 15): Finalization & Documentation

- Refactor and optimize the codebase for reproducibility.
- Prepare the open-source GitHub repository with comprehensive documentation.
- Draft the final project report and create the presentation slides for the QIntern closing events.

Deliverables

1. **Open-Source Repository:** A well-documented public GitHub repository containing the hybrid QML pipelines, QAOA/QRL optimization scripts, and QUBO formulations.
2. **Jupyter Notebooks:** Educational notebooks demonstrating the formulation of energy problems, Quantum Neural ODE construction, and real-hardware error mitigation processes.
3. **Final Report & Presentation:** A detailed PDF report and presentation illustrating the comparative results, noise-resilience analysis, and potential for quantum advantage in energy systems.